Lab: Probability Distribution

Estimated reading time: 30 minutes

In this lab, you will learn how to use the concept of probability distribution in a real-life scenario using Microsoft Excel.

Software used in this lab

The instructional videos in this course use the full Microsoft Excel Desktop version as this has all the available product features, but for the hands-on labs, we will be using the free "Excel for the web" version as this is available to everyone.

Although you can use the Excel Desktop software, it is recommended that you use Excel for the web for the hands-on labs, as the lab instructions specifically refer to this version, and there are some minor differences in the interface and available features.

Data set

This data set has been created manually. Download the following worksheet: M3L1_Lab_Probability_Distribution_Market Return.xlsx.
Note: To download the file, right-click and open the link in a new tab. Then, open the file in Excel for the web to follow the steps given in this lab.

Learning objectives:

- Calculate probabilities from a normal distribution using Excel.
- Calculate the average return and risk based on estimated probability distributions.
- Simulate a continuous probability distribution by sampling values from a discrete distribution to find the average return.

Welcome to the Lab: Probability distribution!

Exercise 1: Calculate Probability Using a Normal Distribution

An electric-bike travels at a top speed of 120 km per hour, whereas the minimum speed is 30 km per hour. Thus, the average speed at which the electric bike travels is 80 km per hour. If the standard deviation is 9, determine the corresponding standard normal distribution (Z-score) for a speed of 90 km per hour.

X is the value for which you want to find the distribution.

Standardized value (X) = 90

Mean (μ) = 80

Standard deviation (σ) = 9

Let's examine the NORM.DIST formula. The NORM.DIST is the Excel version of the Z-table — it automates both the Z-score calculation and the lookup.

= NORM.DIST(x, mean, standard_dev, cumulative)

It takes the actual value (x), mean (μ), and standard deviation (σ), and directly gives the cumulative probability $P(X \leq x)$.

Where:

- x is the value for which you want to find the standard normal distribution
- mean is the given mean
- standard_dev is the standard deviation
- cumulative is a logical function denoting the type of distribution where TRUE outputs the normal distribution function and FALSE outputs the probability density function.

Similarly, the NORM.S.DIST takes the Z-score you already calculated and directly gives the cumulative probability $P(X \leq x)$.

=NORM.S.DIST(z, TRUE)

The problem is solved in Excel, as shown below.

An electric -bike travels at a top speed of 120 Km,
average speed at which the electric bike travels i
the motorbike with a speed more than 90 km/hr.

Standardized Value (X)
Mean (μ)
Standard Deviation (σ)

Calculate the standard normal distribution using

Z = (X - μ) / σ

Calculate the probability using the normal distrib

Cumulative Probability is a logical function denot
FALSE outputs the PROBABILITY DENSIY function.

P(X≤x)= **NORM.DIST(x, mean, standard_dev,TRUE)**

OR

P(X≤x)=

NORM.S.DIST(Z, TRUE)

Answer

Probability of less than X -> =

Probability of greater than X -> (1 - p) =

So, probability of electric -bike travelling with a s
higher than 90 KM PER HR is (in %)

The answer is equal to **0.133**. The probability that any randomly chosen electric bike is moving at more than 90 km per hour is **13.3%**.

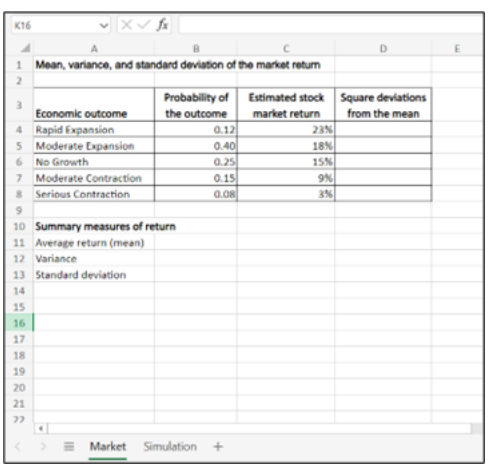
Exercise 2: Estimate the average return and the standard deviation based on the given probability distribution

A stock market investor, Ryan, is concerned about the market returns for the upcoming year. The market return is calculated as a percentage gain over the investment during the year.

Ryan identifies five possible economic outcomes for the country in the coming year: rapid expansion, moderate expansion, no growth, moderate contraction, and serious contraction. He uses the information available to him to assess the probabilities of these economic outcomes and estimate the stock market returns for each case.

Use the provided information to estimate the average stock market return expected and the risk associated with this estimate.

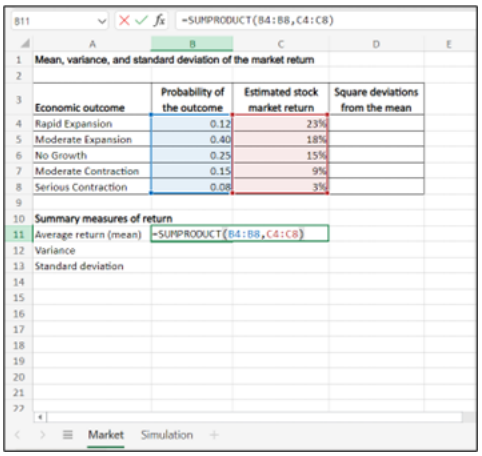
The estimated probabilities and returns are provided in the following worksheet: [M3L1_Lab_Probability_Distribution_Market Return.xlsx](). Refer to the screenshot below.



	A	B	C	D
1	Mean, variance, and standard deviation of the market return			
3	Economic outcome	Probability of the outcome	Estimated stock market return	Square deviations from the mean
4	Rapid Expansion	0.12	23%	
5	Moderate Expansion	0.40	18%	
6	No Growth	0.25	15%	
7	Moderate Contraction	0.15	9%	
8	Serious Contraction	0.08	3%	
10	Summary measures of return			
11	Average return (mean)			
12	Variance			
13	Standard deviation			

1. **Calculate the average return**
   Use the formula: =SUMPRODUCT(array1, [array2], [array3], …)

   In this case, you will calculate the sum product of the probability of the outcome and the estimated return for each outcome, as shown below. You will get an average return is **0.153 or 15.3%**.

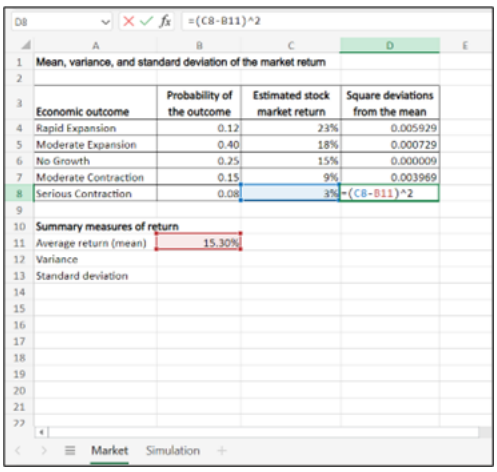


B11 =SUMPRODUCT(B4:B8,C4:C8)

	A	B	C	D	E
1	Mean, variance, and standard deviation of the market return				
2					
3	Economic outcome	Probability of the outcome	Estimated stock market return	Square deviations from the mean	
4	Rapid Expansion	0.12	23%		
5	Moderate Expansion	0.40	18%		
6	No Growth	0.25	15%		
7	Moderate Contraction	0.15	9%		
8	Serious Contraction	0.08	3%		
9					
10	Summary measures of return				
11	Average return (mean)	=SUMPRODUCT(B4:B8,C4:C8)			
12	Variance				
13	Standard deviation				

Market Simulation

2. **Calculate the square deviations from the mean**
   To calculate the overall variance, first calculate the square deviations from the mean. This involves squaring the difference between each individual return and the average return.

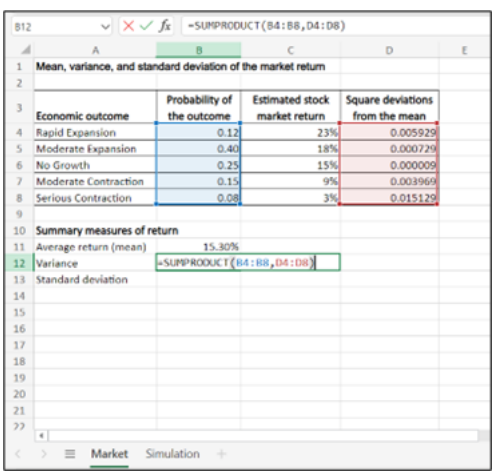


D8 =(C8-B11)^2

	A	B	C	D	E
1	Mean, variance, and standard deviation of the market return				
2					
3	Economic outcome	Probability of the outcome	Estimated stock market return	Square deviations from the mean	
4	Rapid Expansion	0.12	23%	0.005929	
5	Moderate Expansion	0.40	18%	0.000729	
6	No Growth	0.25	15%	0.000009	
7	Moderate Contraction	0.15	9%	0.003969	
8	Serious Contraction	0.08	3%	=(C8-B11)^2	
9					
10	Summary measures of return				
11	Average return (mean)	15.30%			
12	Variance				
13	Standard deviation				

Market Simulation

3. **Calculate the variance**
   Next, calculate the variance. The variance is the sum product of the probability and the square deviations. It will be calculated as shown below. The variance calculated is 0.002811.

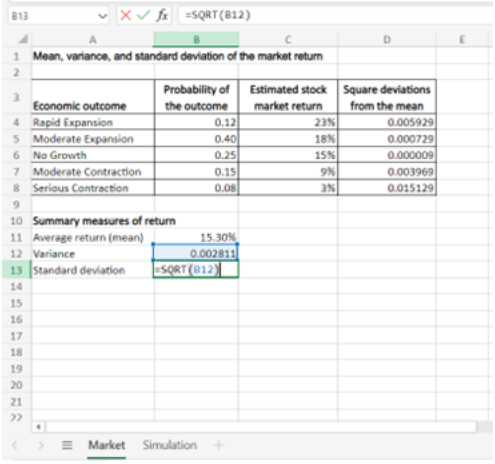


B12 =SUMPRODUCT(B4:B8,D4:D8)

	A	B	C	D	E
1	Mean, variance, and standard deviation of the market return				
2					
3	Economic outcome	Probability of the outcome	Estimated stock market return	Square deviations from the mean	
4	Rapid Expansion	0.12	23%	0.005929	
5	Moderate Expansion	0.40	18%	0.000729	
6	No Growth	0.25	15%	0.000009	
7	Moderate Contraction	0.15	9%	0.003969	
8	Serious Contraction	0.08	3%	0.015129	
9					
10	Summary measures of return				
11	Average return (mean)	15.30%			
12	Variance	=SUMPRODUCT(B4:B8,D4:D8)			
13	Standard deviation				

Market Simulation

4. **Calculate the standard deviation**
   Lastly, calculate the standard deviation. The standard deviation is the square root of the variance. Use the SQRT function to calculate the square root. The answer is 5.3%.



B13 =SQRT(B12)

Mean, variance, and standard deviation of the market return

Economic outcome	Probability of the outcome	Estimated stock market return	Square deviations from the mean
Rapid Expansion	0.12	23%	0.005929
Moderate Expansion	0.40	18%	0.000729
No Growth	0.25	15%	0.000009
Moderate Contraction	0.15	9%	0.003969
Serious Contraction	0.08	3%	0.015129

Summary measures of return

Average return (mean)	15.30%
Variance	0.002811
Standard deviation	=SQRT(B12)

Market Simulation

Exercise 3: Simulate a continuous probability distribution by sampling values from a discrete distribution to find the average return

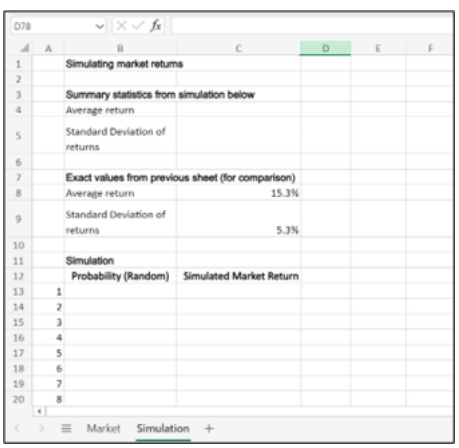
This exercise illustrates how to sample values from a discrete distribution to simulate a continuous probability distribution. Simulate enough economic outcomes and then find the average and standard deviation of the probable return.

To simulate a continuous probability distribution:

- Generate a large number of random probabilities (values from 0 to 1).
- Assign the probable return for each probability based on the estimated values (The estimated stock market return for various economic outcomes and the probability of such outcomes).
- Calculate the mean and standard deviation for the simulated market returns and compare them with the mean and standard deviation originally calculated.

Let's begin. In the Simulation worksheet, populate the average return and standard deviation values calculated in Exercise 1 to help you compare the corresponding values that will be calculated in Exercise 2.

The simulated data set will comprise 400 probabilities randomly generated, and the corresponding market returns based on the research-based estimates provided. Follow the steps to generate the simulated data set.



D78

Simulating market returns

Summary statistics from simulation below

Average return	
Standard Deviation of returns	

Exact values from previous sheet (for comparison)

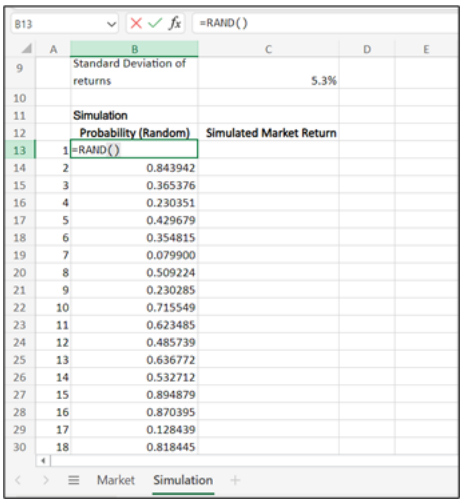
Average return	15.3%
Standard Deviation of returns	5.3%

Simulation

	Probability (Random)	Simulated Market Return
1		
2		
3		
4		
5		
6		
7		
8		

Market Simulation

1. Generate 400 random probability values by typing RAND() in the cell. The RAND() function randomly generates a number between 0 and 1.

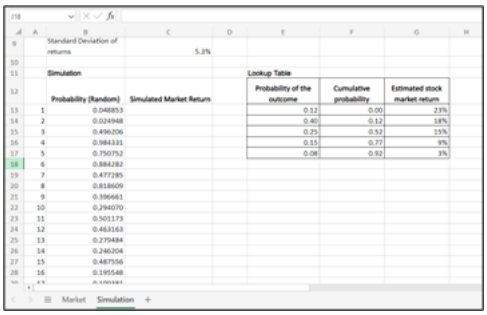


	A	B	C	D	E
9		Standard Deviation of returns	5.3%		
10					
11		Simulation			
12		Probability (Random)	Simulated Market Return		
13	1	=RAND()			
14	2	0.843942			
15	3	0.365376			
16	4	0.230351			
17	5	0.429679			
18	6	0.354815			
19	7	0.079900			
20	8	0.509224			
21	9	0.230285			
22	10	0.715549			
23	11	0.623485			
24	12	0.485739			
25	13	0.636772			
26	14	0.532712			
27	15	0.894879			
28	16	0.870395			
29	17	0.128439			
30	18	0.818445			

2. Next, calculate the simulated market return for each random probability using cumulative probabilities. Cumulative probabilities allow us to determine ranges of probability values to which the market return can be assigned. The table below explains the concept.

Economic outcome	**Probability of the outcome**	**Cumulative probabilities**	**Estimated stock market return**
Rapid Expansion	0.12	0.00	23%
Moderate Expansion	0.40	0.12 (0.00 + 0.12)	18%
No Growth	0.25	0.52 (0.12 + 0.40)	15%
Moderate Contraction	0.15	0.77 (0.52 + 0.25)	9%
Serious Contraction	0.08	0.92 (0.77 + 0.15)	3%

This means that the probability values from 0.00 to 0.12 will be assigned a market return of 23%, probability values from 0.12 to 0.52 will be assigned a market return of 18%, and so on. The probability values from 0.92 to 1 will be assigned a market return of 3%. The market returns will be assigned using the VLOOKUP function. To run VLOOKUP, we need the cumulative probabilities and the market return. Therefore, create the Lookup Table as shown below.



3. Using VLOOKUP, assign market returns to each of the probabilities. Let's examine the VLOOKUP function.

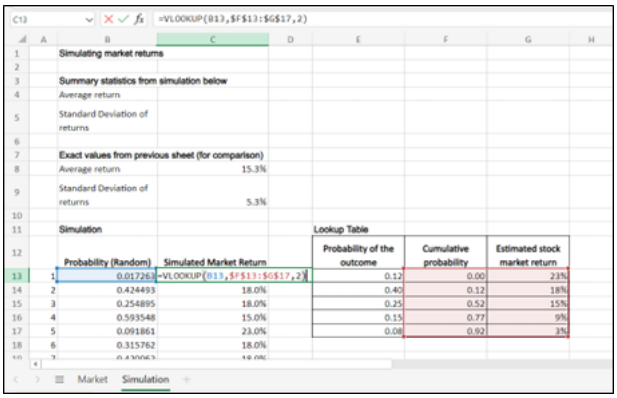
   =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

   lookup_value: The value you want to look up. In this case, it is the random probability value to which you need to assign the market return.

   table_array: The range where the lookup value needs to be searched. Ideally, the first column should have the lookup value. Other columns will have the values that need to be returned. In this case, the first column needs to be the cumulative probabilities, and the second column will be the market return.

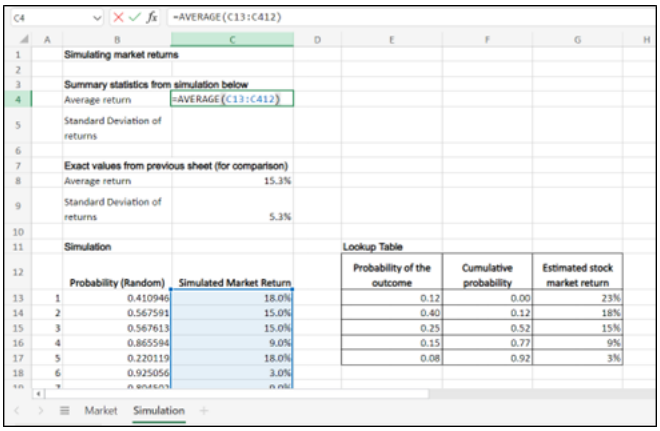
   col_index_num: The column number that has the value that needs to be returned. In this case, it is the second column.

   [range_lookup]: Optionally, you may enter 'TRUE' if you want an approximate match and 'FALSE' if you want an exact match. If no value is entered, the default value is 'TRUE'. In this case, we need an approximate match. Therefore, we can skip entering a value as it will assume the default, 'TRUE'.

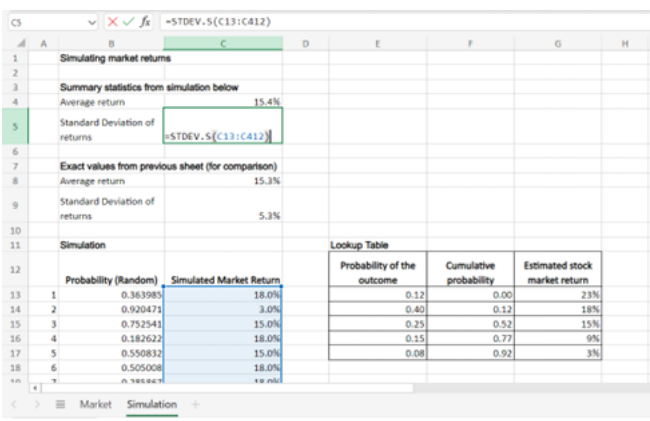


4. Now that the simulated data set has been created, let's calculate the mean and standard deviation of the market return.

   To calculate the mean, use the AVERAGE function. In this case, the mean is 15.5%, which is close to the average return calculated earlier.



To calculate the standard deviation, use the STDEV.S function. The standard deviation is 5.3%, which is very close to the standard deviation calculated earlier.



Congratulations! You've completed the lab activity on probability distribution, and you are ready to move to the next topic.

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